ZHIHENG HUANG (LOWE)

RESEARCH WRITING SAMPLE

Methodology & Results — Selected Dissertation Excerpts

When Ratings and Markets Disagree: TTT-Informed Covariate Models for ATP Tennis Match Prediction

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EXCERPT CONTENTS

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8 consecutive dissertation pages: printed pp. 23–30 (source PDF pp. 36–43)

Logistic regression, feedforward neural networks, validation-based model selection and rolling-origin test evaluation.

02 RESULTS

8 consecutive dissertation pages: printed pp. 40–47 (source PDF pp. 53–60)

Validation, out-of-sample performance, market–TTT gap reduction and directional-disagreement results.

SOURCE AND EDITORIAL SCOPE

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inputs. They provided a reference against which the predictive performance of the TTT baseline and the subsequent covariate models was evaluated.

3.2.2 Logistic Regression Models

Logistic regression was used to examine whether the TTT baseline probability could be improved using additional information available before each match. For match i, the model estimated

$$p_i^{LR} = P(Y_i = 1|x_i)$$

where $Y_i = 1$ indicated that Player A won and x_i denoted the predictors included in the relevant specification.

The TTT baseline probability was first transformed to the log-odds scale:

$$z_i^{TTT} = \log\left(\frac{p_i^{TTT}}{1 - p_i^{TTT}}\right)$$

where p_i^{TTT} is the pre-match probability that Player A wins under the TTT baseline. This transformation allowed the logistic regression to recalibrate the original TTT forecast. Continuous predictors were standardized using means and standard deviations estimated from the training set, with the same transformations then applied to the validation and test sets.

The general logistic regression specification was written as

$$\eta_i = \beta_0 + \beta_1 z_i^{TTT} + \beta_2 u_i^{TTT} + \gamma^\top x_i,$$

where η_i is the linear predictor, β_0 is the intercept, and β_1 measures the contribution of the log-odds TTT prediction. The scalar u_i^{TTT} denotes the uncertainty associated with the TTT skill difference between Players A and B, as defined in Section 3.2.1, while x_i contains the additional pre-match covariates described previously. The coefficient β_2 represents the association between combined TTT uncertainty and the predicted log-odds, while the coefficient vector γ represents the estimated contributions of the additional covariates. The linear predictor was converted into a probability using the logistic function:

$$p_i^{LR} = \frac{1}{1 + exp(-\eta_i)}$$

Equivalently, the model can be expressed on the log-odds scale as

$$\log\left(\frac{p_i^{LR}}{1-p_i^{LR}}\right) = \eta_i.$$

This formulation allows the intercept and the coefficient attached to the TTT prediction to correct systematic bias or overconfidence in the TTT baseline probabilities. The remaining coefficients then determine whether the additional covariates provide predictive information beyond the rating-based estimate.

Eight candidate specifications were fitted, as presented in Table 3.3. The first models examined TTT recalibration and the separate contributions of ranking and historical form. These information groups were then combined in the full covariate model. Short-term rest and workload measures were subsequently added, followed by direct head-to-head and common opponent information, considered separately and jointly.

Table 3.3: Logistic regression candidate specifications

Models	Information
TTT calibration	Log-odds transformation of the original TTT probability
TTT + ranking	TTT probability and uncertainty, rankings, ranking points, and absolute strength gaps
TTT + form	TTT information, career and recent form, surface records, experience, and rest-related information
Full covariates	Combined ranking and form information, together with surface, tournament series, and round
Full + short-term	Full covariate model with additional short-term rest and workload variables
Full + short-term + head-to-head	Short-term model with direct head-to-head information
Full + short-term + common opponent	Short-term model with common opponent information
Full + short-term + head-to-head + common opponent	Short-term model with both relational feature groups

The TTT outputs were used as predictors in a separate logistic regression rather than incorporated into the internal TTT rating process. Bookmaker probabilities were excluded from every specification and retained solely as an external benchmark. Meanwhile, a complete list of variables for each model will be included in the Appendix.

3.2.3 Neural Network Models

A feedforward neural network was used to examine whether a nonlinear mapping of the TTT outputs and additional pre-match covariates could improve the estimated probability of a Player A victory. By allowing nonlinear transformations of the predictors, the network provided a more flexible alternative to the additive linear structure of logistic regression.

The earlier TTT calibration, ranking-only, and form-only specifications were examined using logistic regression as part of the feature group analysis. The neural network analysis therefore did not repeat these preliminary specifications. Instead, the full short-term covariate set was adopted as the common starting point, after which direct head-to-head and common opponent information were evaluated separately and jointly. This design allowed the neural network analysis to examine whether nonlinear modelling added predictive value within the more complete feature sets defined in the logistic regression analysis.

For match i, the input vector was represented as

$$v_i = \begin{bmatrix} z_i^{TTT} \\ u_i^{TTT} \\ c_i \end{bmatrix},$$

where z_i^{TTT} denotes the log-odds transformation of the TTT baseline win probability, u_i^{TTT} contains the combined TTT uncertainty, and c_i contains the additional pre-match covariates included in the relevant specification.

Let P denote the number of input variables and H the number of units in the hidden layer. For hidden unit j, the activation for match i can be written as

$$h_{ij} = g\left(b_j + \sum_{r=1}^{P} w_{jr}\, v_{ir}\right), \quad j = 1, \ldots, H,$$

where v_{ir} is the value of input r, w_{jr} is the weight connecting that input to hidden unit j, and b_j is the corresponding bias term. The function $g(\cdot)$ is the logistic activation function,

$$g(a) = \frac{1}{1 + exp(-a)}.$$

By passing a weighted combination of the inputs through this nonlinear transformation,

the network can represent relationships and interactions that do not need to be specified individually in the model formula.

The activations from the hidden layer were then combined in a single output unit. The output layer linear predictor was defined as

$$\eta_i^{NN} = b_o + \sum_{j=1}^{H} a_j \, h_{ij},$$

where a_j is the weight connecting hidden unit j to the output unit and b_o is the output bias. A logistic output function converted this value into the predicted probability that Player A wins match i:

$$p_i^{NN} = \frac{1}{1 + exp\left(-\eta_i^{NN}\right)}.$$

The resulting prediction is bounded between zero and one and is therefore directly comparable with the TTT baseline and bookmaker probabilities.

The network parameters were estimated using entropy fitting, corresponding to binary cross-entropy for the observed match outcome. Weight decay was included to discourage excessively large network weights and reduce the risk of fitting highly irregular patterns in the training data. The penalized objective can be represented as

$$\mathcal{J}(\theta) = -\sum_{i=1}^{n_{\text{train}}} \left[Y_i \log\left(p_i^{NN}\right) + (1 - Y_i) \log\left(1 - p_i^{NN}\right)\right] + \lambda \sum_{k=1}^{K} w_k^2,$$

where θ denotes the fitted network parameters, θ_j denotes a penalized network weight, and λ is the weight decay parameter. The first component penalizes probability estimates that assign low probability to the observed result, while the second restricts the overall magnitude of the fitted weights. Network optimization used the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm. The initial comparison of the four feature families used a maximum of 300 iterations. Because the selected short-term family showed nonconvergence at this limit, its complete 48-configuration validation grid and all 876 rolling test fits were rerun with a maximum of 1,000 iterations for the final primary analysis. All final short-term validation and test fits converged.

Four neural network families were fitted, as listed in Table 3.4. The short-term specification formed the common base and included the TTT outputs, the main pre-match covariates, and the short-term workload and rest variables. The remaining

specifications added direct head-to-head information, common opponent information, or both groups of relational covariates.

Table 3.4: Neural network candidate specifications

Models	Information
Short-term	TTT outputs, main pre-match covariates, and short-term workload and rest variables
Short-term + head-to-head	Short-term specification with direct head-to-head variables
Short-term + common opponent	Short-term specification with common opponent variables
Short-term + head-to-head + common opponent	Short-term specification with both head-to-head and common opponent variables

Before estimation, the categorical variables for playing surface, tournament series, and tournament round were converted into indicator columns through a design matrix. The intercept column produced during this conversion was removed because the network contained separate bias parameters.

Hyperparameter selection was conducted separately for each of the four feature specifications. The number of hidden units was varied across 3, 5, 8, and 12, while the candidate weight decay values were 0.0001, 0.001, 0.01, and 0.05. Network estimation depends on the randomly generated initial weights, so each combination of hidden layer size and weight decay was fitted using three random seeds: 101, 202, and 303. This produced 48 candidate fits for each feature specification and 192 candidate fits across the four neural network families.

Each candidate network was fitted using the training sample and evaluated on the validation sample. Within each feature specification, the combination of hidden layer size, weight decay, and random seed that produced the lowest validation log loss was retained. The selected configurations are reported in Table 3.5.

Table 3.5: Neural network configurations selected by validation log loss

Models	Hidden units	Weight decay	Random seed
Short-term	3	0.0100	303
Short-term + head-to-head	3	0.0100	202
Short-term + common opponent	3	0.0500	101
Short-term + head-to-head + common	3	0.0001	101

Models	Hidden units	Weight decay	Random seed
opponent			

The primary short-term model was selected from the fully converged 1,000-iteration validation rerun, whereas the three extensions retained the configurations selected in the initial analysis. The random seed identifies only the initialization that produced the lowest validation log loss and should not be interpreted as a substantive model parameter.

3.3 Model Selection and Testing

The three chronological periods served different roles in model development and evaluation, as presented in Table 3.6. Matches from 2010 to 2021 were used to fit the candidate logistic and neural network models and to estimate the required imputation and standardization values. Matches from 2022 were used to compare model specifications and select the neural network hyperparameters. After the feature sets and hyperparameters had been fixed, performance was evaluated sequentially over 2023–2025 using the rolling-origin procedure described in Section 3.3.2. Before forecasting the matches on each test date, model parameters were re-estimated using all matches observed before that date, including earlier matches from the 2023–2025 evaluation period. Once the outcomes from a test date became available, those matches were added to the historical data used for subsequent forecasts. Test-period outcomes were therefore used only after they had occurred and were not used for feature selection or hyperparameter tuning.

Table 3.6: Roles of the chronological data partitions

Data partition	**Role in the modelling framework**
Training set	Fit candidate models, estimate preprocessing values, and contribute to every test-date estimation history
Validation set	Compare model specifications and neural network configurations; completed validation matches later enter the test-date estimation history
Test set	Evaluate sequential forecasts; after each date, completed results enter the history for later test dates

3.3.1 Model Selection

Validation log loss was used as the principal criterion for comparing candidate model

specifications, calculation of which is about to be defined in Section 3.4.1. For a candidate set $\mathcal{M}$, the preferred model was defined as

$$m^* = \arg\min_{m \in \mathcal{M}} \text{LogLoss}_{\text{val}}(m),$$

where $\text{LogLoss}_{\text{val}}(m)$ denotes the log loss achieved by model m on the validation set. Brier score and classification accuracy were also calculated during validation, but these measures were used to provide complementary information rather than to determine the primary model.

For logistic regression, the eight specifications described in Section 3.2.2 were fitted separately using the training data. Their predictor sets were defined before the validation comparison, and no numerical hyperparameter tuning was required. The validation stage therefore assessed how predictive performance changed across the alternative combinations of TTT outputs, ranking variables, form measures, short-term workload information, direct head-to-head records, and common opponent features. Under the validation log-loss criterion, the full short-term specification with direct head-to-head variables was selected as the primary logistic regression model. The remaining specifications were retained for comparison in order to examine the contribution of the different feature groups.

The neural network selection procedure involved two stages. First, within each of the four feature families described in Section 3.2.3, candidate combinations of hidden layer size, weight decay, and random initialization were fitted using the training data. The configuration with the lowest validation log loss was retained within each feature family. Second, the four retained family-level networks were compared using the same validation criterion to identify the primary neural network specification.

The random seeds represented alternative initial values for the network weights rather than differences in the substantive information supplied to the model. Comparing several initializations reduced the possibility that a feature family would be evaluated on the basis of an unusually poor fitted solution. Following the two stage validation procedure, the short-term neural network without the additional head-to-head or common opponent extensions achieved the lowest validation log loss and was therefore selected as the primary neural network model. The head-to-head, common opponent, and combined specifications were retained as predefined extensions for comparison on the test set.

3.3.2 Rolling Origin Test Evaluation

By the end of validation, the predictor sets, the identities of the primary models, and the neural network hyperparameters had been fixed. The imputation and standardization values estimated from the initial training period were also kept unchanged. None of these choices was reconsidered using results from 2023–2025. The fitted logistic regression coefficients and neural network weights, however, were not fixed. Before each distinct test date, every predefined logistic regression and neural network model was re-estimated using an expanding history containing only matches completed before that date.

For a test date d, the available estimation history was defined as

$$H_d = \{j : t_j < d\}$$

where t_j denotes the date of match j. For model m, the parameters and the prediction for test match i were obtained as

$$\hat{\theta}_{m,d} = Fit_m\left(\{(\mathbf{x}_j, y_j) : j \in H_d\}\right), \quad \hat{p}_{im} = f_m\left(\mathbf{x}_i; \hat{\theta}_{m,d}\right), \quad t_i = d.$$

where $\mathbf{x}_j$ and y_j denote the pre-match predictors and observed outcome for historical match j, respectively. The term $\hat{\theta}_{m,d}$ represents the parameters estimated from the history available before date d, while $\hat{p}_{im}$ is the predicted probability that Player A wins match i.

As all matches played on the same date were forecast using the same historical information, and their outcomes were added to the estimation history only after predictions had been generated for every match on that date. The first test-date fit therefore used all matches from the training and validation periods. Later fits also included completed matches from earlier dates within the 2023–2025 evaluation period. Consequently, the outcome of a test match could not influence its own prediction or another prediction made on the same date, but it could contribute to parameter estimation for later dates. The selected model specifications and neural network hyperparameters remained unchanged throughout this procedure.

Although a primary logistic regression model and a primary neural network model had been selected via validation using the log loss, test predictions were nevertheless generated for all predefined specifications. These results enable a comparison of the contributions of different feature sets and an examination of the sensitivity of the

Chapter 4

Results

This chapter presents the empirical results of the forecasting analysis. It first reports the validation performance used to select the primary logistic regression and neural network specifications. The out-of-sample testing performance of these models is then compared with the TTT benchmark and the bookmakers' benchmark, and an analysis is conducted to determine the extent to which the covariate models can bridge the performance gap between TTT and the bookmakers. The chapter subsequently examines direction disagreements between TTT and the bookmaker and evaluates how disagreement frequency and prediction error vary across levels of TTT uncertainty. Lastly, the main findings are summarized at the end of the chapter.

4.1 Model Selection

This section presents the validation performance used to identify the primary logistic regression and neural network specifications. As described in previous chapter, validation log loss was the model selection criterion, while the Brier score and classification accuracy were reported as complementary measures.

4.1.1 Logistic Regression Models

Table 4.1 presents the validation performance of the eight logistic regression specifications. Validation log loss was used as the model-selection criterion, while the Brier score and classification accuracy were included as complementary measures.

Table 4.1: Validation performance of the logistic regression specifications

Model	Log loss	Brier score	Accuracy
TTT calibration	0.6248	0.2172	0.6455
TTT + ranking	0.6182	0.2147	0.6522
TTT + form	0.6194	0.2152	0.6483
Full covariates	0.6094	0.2112	0.6593
Full + short-term	0.6092	0.2110	0.6597
Full + short-term + head-to-head	**0.6084**	**0.2107**	**0.6605**
Full + short-term + common opponent	0.6095	0.2112	0.6562
Full + short-term + head-to-head + common opponent	0.6087	0.2109	0.6577

The TTT calibration specification recorded a validation log loss of 0.6248. The ranking and form specifications produced lower values of 0.6182 and 0.6194, respectively, while the full covariate specification reduced the value further to 0.6094. The addition of short-term variables resulted in a log loss of 0.6092. Among the extended specifications, the model including head-to-head information recorded the lowest validation log loss of 0.6084. It also had the lowest Brier score, at 0.2107, and the highest classification accuracy, at 0.6605. This specification was therefore selected as the primary logistic regression model. The common-opponent and combined head-to-head and common-opponent specifications recorded validation log losses of 0.6095 and 0.6087, respectively.

4.1.2 Neural Network Models

Table 4.2 presents the validation performance of the selected configuration within each neural network feature family. The final short-term primary configuration was selected from its fully converged 1,000-iteration rerun, while the three predefined extension configurations were retained from the original family-specific validation comparisons.

Table 4.2: Validation performance of the selected neural network specifications

Model	Log loss	Brier score	Accuracy
Short-term	0.6038	0.2090	0.6621
Short-term + head-to-head	0.6078	0.2105	0.6530
Short-term + common opponent	0.6071	0.2104	0.6506

Model	Log loss	Brier score	Accuracy
Short-term + head-to-head + common opponent	0.6050	0.2094	0.6570

The short-term specification recorded the lowest validation log loss of 0.6038. It also had the lowest Brier score, at 0.2090, and the highest classification accuracy, at 0.6621. It was therefore selected as the primary neural network model. The combined head-to-head and common-opponent specification recorded the second-lowest validation log loss of 0.6050. The common-opponent and head-to-head specifications produced validation log losses of 0.6071 and 0.6078, respectively.

4.2 Overall Test Performance

4.2.1 Primary Models and Benchmarks

Following the validation-stage selection, the primary logistic regression and neural network models were evaluated on the test set alongside the TTT baseline and bookmaker benchmark. Table 4.3 reports their performance across the 7,725 matches played between 2023 and 2025.

Table 4.3: Test performance of the primary models and benchmarks over 2023-2025

Forecast	Specification	Logloss	Brierscore	Accuracy	Meanprobability
Bookmaker benchmark	—	0.5866	0.2019	68.13%	0.5927
TTT baseline	—	0.6576	0.2275	63.70%	0.5753
Primary logistic regression	Full + short-term + head-to-head	0.6169	0.2148	65.48%	0.5768
Primary neural network	Short-term	0.6159	0.2143	64.96%	0.5783

The bookmaker benchmark recorded the lowest test log loss of 0.5866 and the lowest Brier score of 0.2019. It also produced the highest classification accuracy, at 68.13%, and the highest mean probability assigned to the actual winner, at 0.5927.

The TTT baseline recorded a log loss of 0.6576, a Brier score of 0.2275, and an accuracy of 63.70%. Both primary covariate models recorded lower log loss and Brier score values than the TTT baseline, together with higher classification accuracy. The primary neural network recorded a log loss of 0.6159, a Brier score of 0.2143, and an

accuracy of 64.96%. The primary logistic regression model recorded corresponding values of 0.6169, 0.2148, and 65.48%. The neural network therefore had slightly lower probability error, while logistic regression had slightly higher accuracy. Their mean probabilities assigned to the actual winner were 0.5783 and 0.5768, respectively.

4.2.2 Other Predefined Specifications

The remaining predefined logistic regression and neural network specifications were also evaluated on the same test set. Table 4.4 reports their test performance.

Table 4.4: Test performance of the other predefined specifications

Model	Log loss	Brier score	Accuracy	Mean probability
Logistic: full + short-term	0.6175	0.2151	65.31%	0.5765
Logistic: full + short-term + common opponent	0.6176	0.2151	65.39%	0.5765
Logistic: full + short-term + head-to-head + common opponent	0.6170	0.2148	65.28%	0.5768
Neural network: short-term + head-to-head	0.6168	0.2145	65.63%	0.5783
Neural network: short-term + common opponent	0.6143	0.2138	65.17%	0.5791
Neural network: short-term + head-to-head + common opponent	0.6166	0.2149	65.10%	0.5786

The three logistic regression specifications produced similar test results. Their log loss ranged from 0.6170 to 0.6176, while their classification accuracy ranged from 65.28% to 65.39%. The combined head-to-head and common-opponent specification recorded the lowest log loss among these three models, at 0.6170.

Among the neural network extensions, the common-opponent specification recorded the lowest observed test log loss, at 0.6143, and the lowest Brier score, at 0.2138. The head-to-head specification had the highest classification accuracy, at 65.63%, while the head-to-head and combined specifications recorded log losses of 0.6168 and 0.6166, respectively. Although the common-opponent extension had a lower observed test log loss than the primary short-term neural network, it remained an extension because the primary specification had been selected using validation performance.

4.3 Log-loss Gap Reduction

The test-set log-loss difference between the TTT baseline and the bookmaker benchmark was 0.0709, based on log-loss values of 0.6576 and 0.5866, respectively. Figure 4.1 presents the proportion of this gap closed by the short-term logistic regression and neural network specifications and their predefined extensions.

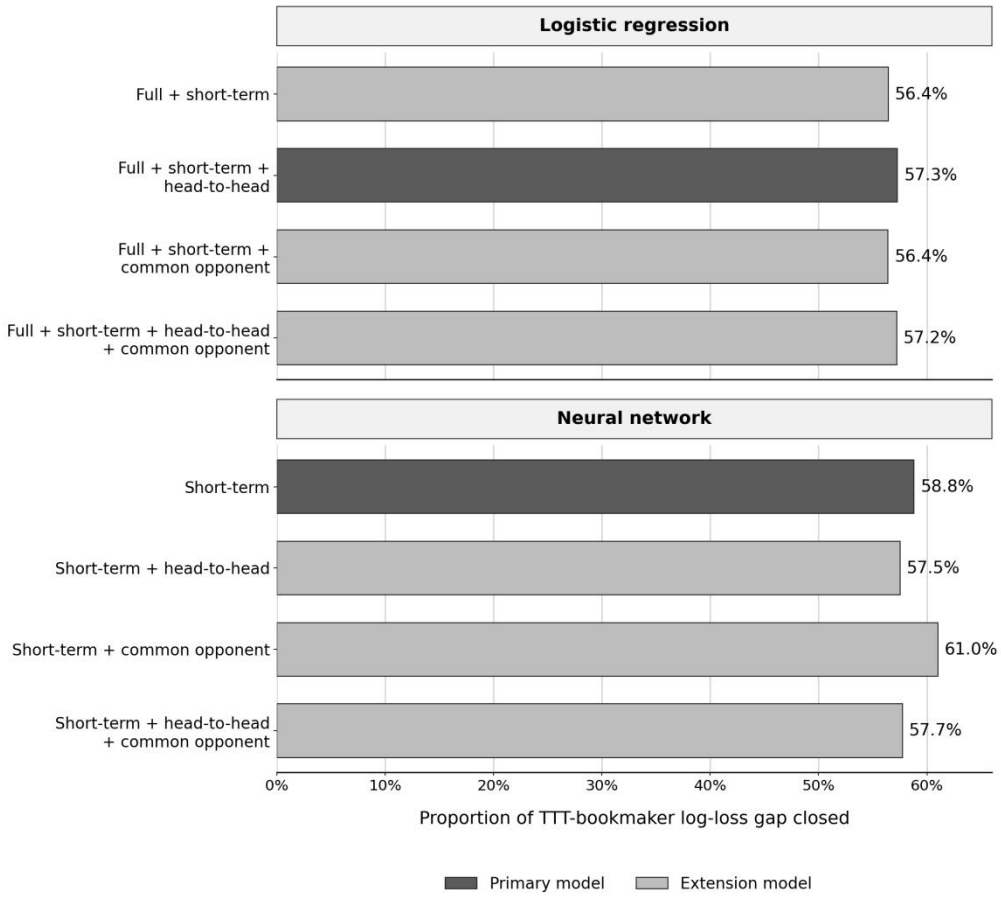


Figure 4.1: Proportion of the test set TTT − bookmaker log loss gap closed by the covariate models. Dark bars indicate the primary models selected by validation performance.

The primary logistic regression model closed 57.3% of the test-set log-loss gap, while the primary neural network closed 58.8%. The difference between the two primary models was 1.5 percentage points. Across the logistic regression specifications, gap reduction ranged from 56.4% to 57.3%. The neural network specifications ranged from 57.5% to 61.0%. Among the predefined extensions, the common-opponent neural network recorded the highest gap reduction, at 61.0%. The primary model designations remained those determined from validation performance.

4.4 Direction Disagreement

4.4.1 Frequency

A direction disagreement occurred when the TTT baseline and bookmaker benchmark assigned probabilities above 0.5 to different players. Table 4.5 reports the number and proportion of disagreement matches in the full sample and each chronological partition.

Table 4.5: Frequency of TTT–bookmaker direction disagreement

Sample	Disagreement matches	Total matches	Disagreement rate
Full sample	7,145	39,154	18.25%
Training set	5,208	28,890	18.03%
Validation set	441	2,539	17.37%
Test set	1,496	7,725	19.37%

Across the full analytical sample, the TTT baseline and bookmaker benchmark predicted different winners in 7,145 of 39,154 matches, corresponding to a disagreement rate of 18.25%. The disagreement rate was 18.03% in the training set and 17.37% in the validation set. In the test set, 1,496 of 7,725 matches produced a direction disagreement, giving the highest partition-specific rate of 19.37%.

4.4.2 Test-set Disagreement Outcomes

Among the 1,496 test matches in which the two forecasts disagreed, exactly one of the bookmaker and TTT baseline directions was correct. Table 4.6 reports the corresponding outcomes.

Table 4.6: Outcomes of test matches that TTT and the bookmaker predicted differently

Outcome	Matches	Proportion
Bookmaker correct, TTT baseline incorrect	919	61.43%
TTT baseline correct, bookmaker incorrect	577	38.57%
Total	1,496	100.00%

The bookmaker benchmark predicted the correct winner in 919 disagreement matches, representing 61.43% of the test disagreement subset. The TTT baseline predicted the correct winner in the remaining 577 matches, corresponding to 38.57%.

4.5 Performance in Disagreement Matches

The predictive metrics were recalculated for the 1,496 test matches in which the TTT baseline and bookmaker benchmark predicted different winners. Table 4.7 compares the two benchmarks, the primary models, and the predefined logistic regression and neural network extensions within this subset.

Table 4.7: Predictive performance within the test-set disagreement subset

Model	Log loss	Brier score	Accuracy	Mean probability
Bookmaker benchmark	0.6553	0.2318	61.43%	0.5345
TTT baseline	0.8239	0.3031	38.57%	0.4683
Logistic: full + short-term	0.7088	0.2571	51.34%	0.5048
Logistic: full + short-term + head-to-head (primary)	0.7065	0.2560	52.01%	0.5058
Logistic: full + short-term + head-to-head + common opponent	0.7060	0.2558	51.40%	0.5061
Neural network: short-term (primary)	0.6973	0.2517	51.20%	0.5118
Neural network: short-term + head-to-head	0.7015	0.2538	52.34%	0.5105
Neural network: short-term + common opponent	0.6995	0.2530	50.94%	0.5115
Neural network: short-term + head-to-head + common opponent	0.7023	0.2540	52.34%	0.5094

The bookmaker benchmark recorded the lowest log loss of 0.6553 and the lowest Brier score of 0.2318. Its classification accuracy was 61.43%, compared with 38.57% for the TTT baseline. The TTT baseline also recorded the highest log loss, at 0.8239, and the highest Brier score, at 0.3031.

The primary logistic regression model reduced log loss to 0.7065, while the primary neural network recorded a lower value of 0.6973. Their Brier scores were 0.2560 and 0.2517, and their accuracies were 52.01% and 51.20%, respectively. The primary neural network recorded the lowest log loss and Brier score among the fitted covariate models. The head-to-head and combined neural network extensions recorded the highest accuracy, both at 52.34%.

The log-loss gap between the TTT baseline and bookmaker benchmark within the disagreement subset was 0.1686. The primary logistic regression model closed 69.6% of

this gap, while the primary neural network closed 75.1%. Among the neural network extensions, the common-opponent specification closed 73.7%, the head-to-head specification closed 72.6%, and the combined specification closed 72.1%. The primary neural network therefore recorded the largest gap reduction among the fitted covariate models in this subset.

4.6 TTT Uncertainty

4.6.1 Disagreement Rate

Figure 4.2 presents the proportion of test matches producing a TTT–bookmaker direction disagreement within each TTT uncertainty group. The group boundaries were derived from the training-set uncertainty distribution, as described in Section 3.4.3.

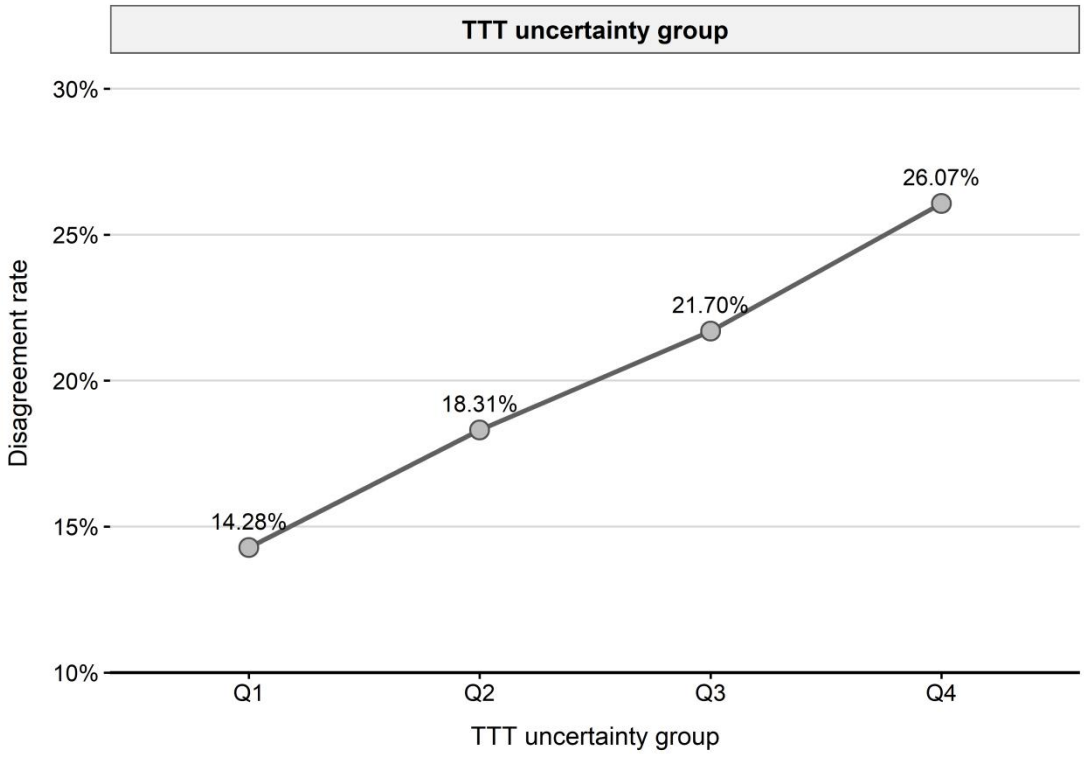


Figure 4.2: TTT – bookmaker disagreement rate across TTT uncertainty groups in the test set. Q1 represents the lowest uncertainty and Q4 the highest.

The disagreement rate increased across the four uncertainty groups. It was 14.28% in Q1, rising to 18.31% in Q2 and 21.70% in Q3. The highest rate was recorded in Q4, where 26.07% of test matches produced different predicted winners from the TTT baseline and bookmaker benchmark. The difference between Q1 and Q4 was 11.79